

# PS200C: Quantitative Methods

## In-class Activity 8: Instrumental Variables

Name: \_\_\_\_\_

**Setup for Q1–Q2.** Suppose a city is studying whether being incarcerated (vs. a non-custodial sentence) for a first offense affects re-arrest within five years. In this jurisdiction there are two judges who hear first-offense cases: **Judge A** (relatively lenient) and **Judge B** (relatively stringent). Defendants are assigned (essentially at random) to one of the two.

The researcher proposes:

- $Z = 1$  if assigned to Judge B (stringent);  $Z = 0$  if assigned to Judge A (lenient).
- $D = 1$  if incarcerated;  $D = 0$  if non-custodial sentence.
- $Y = 1$  if re-arrested within five years.

**Question 1: Who are the compliers?** In one sentence, describe the compliers in this design — the subgroup whose treatment status would be moved by the instrument.

**Question 2: State the exclusion restriction.** In one sentence, write what the exclusion restriction *requires* here. Can you think of anything that might violate it in this setting?

**Question 3: Understanding the assumptions.** A researcher proposes using *distance to the nearest college* as an instrument for years of education, with adult earnings as the outcome.

- How might the ignorability assumption be violated here?

- How might the exclusion restriction be violated here?

**Question 4: Wald by hand.** Suppose you have a randomized encouragement-design experiment.  $Z = 1$  if assigned to receive a free flu-shot voucher;  $D = 1$  if the person actually got vaccinated;  $Y$  is days of missed work that flu season.

|           | $Z = 0$ (no voucher) | $Z = 1$ (voucher) |
|-----------|----------------------|-------------------|
| $\bar{D}$ | 0.20                 | 0.55              |
| $\bar{Y}$ | 3.0 days             | 2.3 days          |

(a) Compute the ITT (effect of  $Z$  on  $Y$ ) and the first stage (effect of  $Z$  on  $D$ ).

(b) Compute the Wald / IV estimate for the effect of vaccination on missed work.

(c) For whom is this the treatment effect? (Describe in a sentence)